

Analogue Electronics – 1. Sheet

Designing active Filters – an Example

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In this application note, a butterworth bandpass-filter is designed. The tolerance-scheme is shown in figure 1.

1 Design of a Bandpass-Filter

First, some symbols are introduced. All the normalized

Center frequency

$$f_C = \sqrt{f_L \cdot f_U} \quad (1)$$

normalized frequency of the bandpass filter

$$\omega_{n,BP} = \frac{\omega}{\omega_C} \quad (2)$$

bandwidth of the bandpass filter

$$\omega_{n,BP,BW} = \frac{\Delta\omega}{\omega_C} \quad (3)$$

the complex frequency variable, which is also normalized to the center frequency of the bandpass filter by

$$s_{n,BP} = \frac{s}{\omega_C} \quad (4)$$

For the actual problem, the upper corner frequency is

$$f_U = \frac{f_C^2}{f_L} = 125 \text{ kHz} \quad (5)$$

The bandwidth is calculated to

$$\Delta f = 125 \text{ kHz} - 80 \text{ kHz} = 45 \text{ kHz}, \quad (6)$$

which is normalized to

$$\omega_{n,BP,BW} = \frac{2\pi \cdot 45 \text{ kHz}}{2\pi \cdot 100 \text{ kHz}} = 0.45. \quad (7)$$

The lower normalized corner frequency is

$$\omega_{n,BP,L} = \frac{\omega_L}{\omega_C} = 0.8, \quad (8)$$



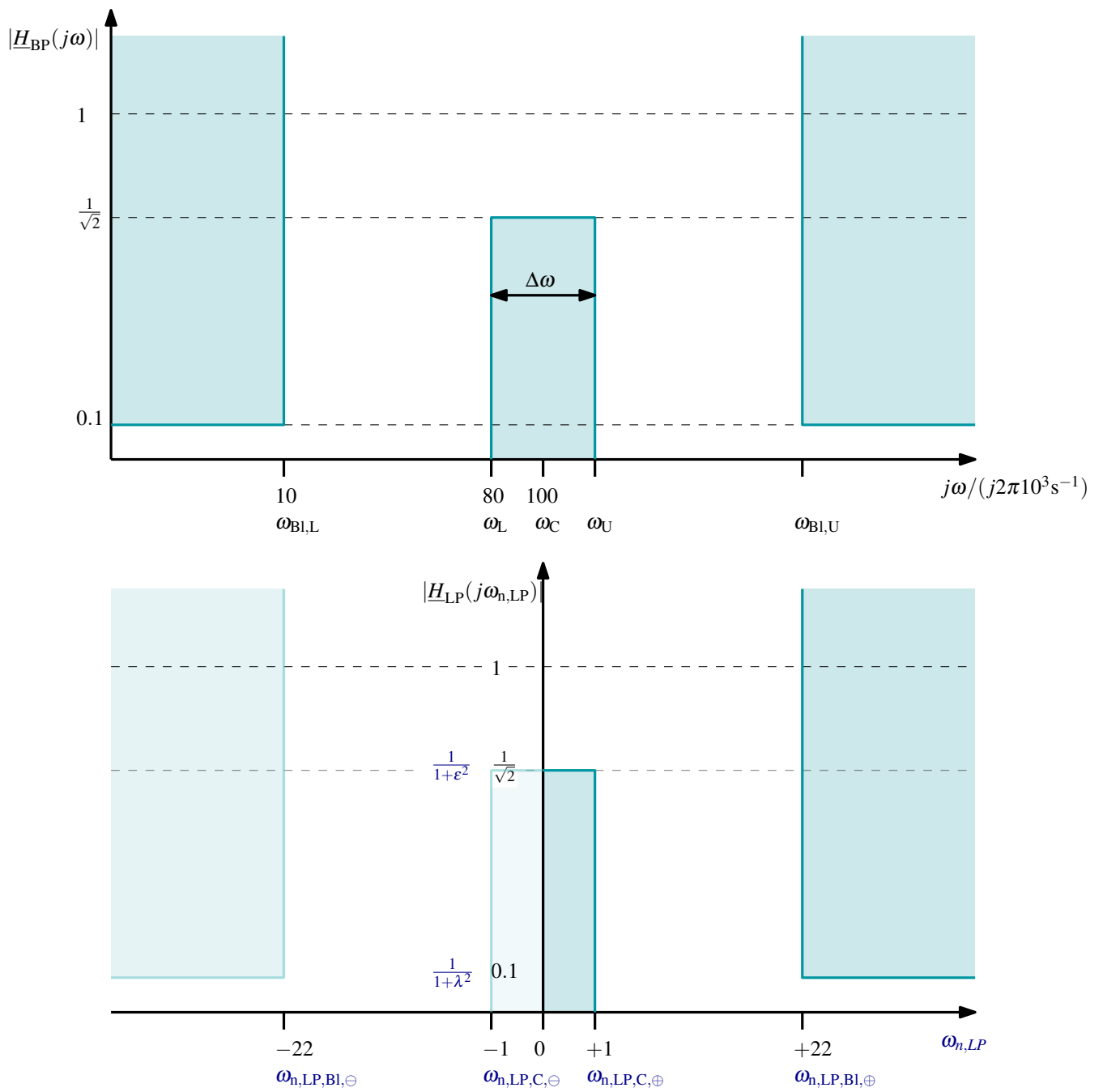


Fig. 1: Bandpass and corresponding Lowpass Tolerance Scheme

while the upper corner frequency is

$$\omega_{n,BP,U} = \frac{\omega_U}{\omega_C} = 1.25. \quad (9)$$

The upper blocking frequency is calculated by using the known lower one and their relation to the normalized center frequency $\omega_{n,BP,C}$ by using

$$\omega_{n,BP,C} = \sqrt{\omega_{n,BP,BL,L} \cdot \omega_{n,BP,BL,U}} \quad (10)$$

yields

$$\omega_{n,BP,BL,U} = \frac{\omega_{n,BP,BL,L}}{\omega_{n,BP,C}} \quad (11)$$

The next step is the transformation of the bandpass filter into a corresponding lowpass filter. All the lowpass values are blue for better understanding. The transformation is

$$s_{n,LP} = \frac{1}{\omega_{n,BP,BW}} \left(s_{n,BP} + \frac{1}{s_{n,BP}} \right) \quad (12)$$

and for the normalized frequencies

$$j\omega_{n,LP} = \frac{1}{\omega_{n,BP,BW}} \left(j\omega_{n,BP} + \frac{1}{j\omega_{n,BP}} \right) \quad (13)$$

$$= j \frac{1}{\omega_{n,BP,BW}} \left(\omega_{n,BP} - \frac{1}{\omega_{n,BP}} \right) \quad (14)$$

Using the transformation (14), some characteristic points can be transformed from the bandpass into its corresponding low-pass points. The normalized upper corner frequency $\omega_{n,BP,U}$ has the corresponding low pass corner frequency in the negative frequency domain

$$\omega_{n,LP,C\ominus} = \frac{1}{\omega_{n,BP,BW}} \left(\omega_{n,BP,L} - \frac{1}{\omega_{n,BP,L}} \right) \quad (15)$$

$$\omega_{n,LP,C\ominus} = \frac{1}{0.45} \left(0.8 - \frac{1}{0.8} \right) = -1 \quad (16)$$

The other characteristic points are

$$\omega_{n,BP,U} \rightarrow \omega_{n,LP,C\oplus} = +1 \quad (17)$$

$$\omega_{n,BP,BL,L} \rightarrow \omega_{n,LP,BL\ominus} = -22 \quad (18)$$

$$\omega_{n,BP,BL,U} \rightarrow \omega_{n,LP,BL\oplus} = +22 \quad (19)$$

$$(20)$$

The degree or the corresponding lowpass can be calculated by

$$n \geq \frac{\log \frac{\lambda}{\varepsilon}}{\log \frac{\omega_{n,LP,BL}}{\omega_{n,LP,C}}} \quad (21)$$

here λ and ε are

$$\varepsilon = \sqrt{\frac{1}{|H(j\omega)|^2} - 1} = \sqrt{\frac{1}{\left(\frac{1}{\sqrt{2}}\right)^2} - 1} = 1 \quad (22)$$

$$\lambda = \sqrt{\frac{1}{|H(j\omega)|^2} - 1} = \sqrt{\frac{1}{|0.1|^2} - 1} = \sqrt{99} \quad (23)$$



so the degree of the low pass filter will be calculated to

$$n \geq \frac{\log \frac{\sqrt{99}}{1}}{\log 22} = 0.743. \quad (24)$$

Thus, a first order butterworth lowpass filter will be sufficient, its transfer function is

$$\underline{H}_{LP} = \frac{1}{1 + s_{n,LP}} \quad (25)$$

This transfer function has to be inverse transformed back into the bandpass domain. The transformation (12) is inserted in the transfer function (25)

$$\underline{H}_{BP}(s_{n,BP}) = \frac{1}{1 + \frac{1}{\omega_{n,BP,BW}} \left(s_{n,BP} + \frac{1}{s_{n,BP}} \right)}, \quad (26)$$

which will be denormalized by inserting

$$s_{n,BP} = \frac{s}{\omega_C} \quad (27)$$

$$\omega_{n,BP,BW} = \frac{\Delta\omega}{\omega_C} \quad (28)$$

resulting in the transfer function of the bandpass filter

$$\underline{H}_{BP}(s) = \frac{\frac{\Delta\omega}{\omega_C} \cdot \frac{s}{\omega_C}}{1 + \frac{\Delta\omega}{\omega_C^2} s + \frac{1}{\omega_C^2} s^2} \quad (29)$$

$$\underline{H}_{BP}(s) = \frac{\frac{\Delta\omega}{\omega_C^2} s}{1 + \frac{\Delta\omega}{\omega_C^2} s + \frac{1}{\omega_C^2} s^2} \quad (30)$$

$$\underline{H}_{BP}(s) = \frac{a_1 s}{1 + b_1 s + b_2 s^2}. \quad (31)$$

So, a second order structure will be a valid solution, the coefficients are

$$a_1 = b_1 = \frac{\Delta\Omega}{\omega_C^2} \approx 7.162 \cdot 10^{-7} \text{sec} \quad (32)$$

$$b_2 = \frac{1}{\omega_C^2} \approx 2.533 \cdot 10^{-12} \text{sec}^2 \quad (33)$$

2 Realisation of the Filter

2.1 Complex Transfer Function

The first step is to calculate the complex transfer function $\underline{H}_{MFL}(s)$. This will be done in general, so all the elements are complex impedances. The application lateron will help to decide, which component is selected to be a capacitor and which is a resistor. To reduce the count of fractions, it is very helpful to use complex conductances.

The node equations are for node l

$$0 = \underline{I}_1 + \underline{I}_2 - \underline{I}_3 \quad (34)$$

$$0 = (\underline{V}_E - \underline{V}_3)\underline{Y}_1 + (-\underline{V}_A - \underline{V}_3)\underline{Y}_2 - \underline{V}_3\underline{Y}_3 - \underline{V}_3\underline{Y}_4 \quad (35)$$



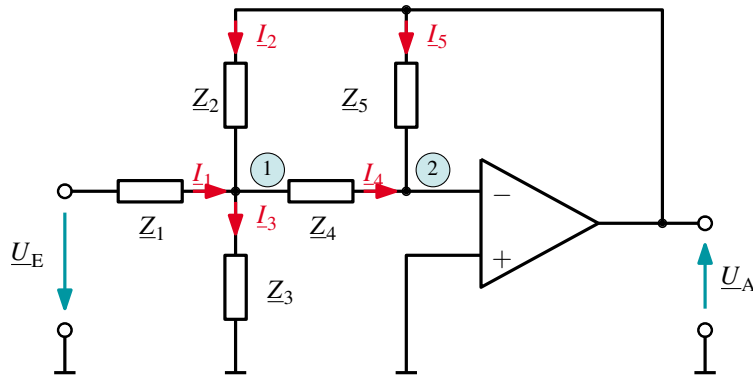


Fig. 2: Multiple Feedback Loop Structure using complex elements

and for node 2

$$0 = I_4 - I_5 \quad (36)$$

$$0 = \underline{V}_3 \underline{Y}_4 - \underline{U}_A \underline{Y}_5 \quad (37)$$

which is rearranged to get $\underline{V}_3$

$$\underline{V}_3 = \underline{V}_A \frac{\underline{Y}_5}{\underline{Y}_4}. \quad (38)$$

This voltage will be inserted in (35)

$$0 = \left(\underline{V}_E - \underline{V}_A \frac{\underline{Y}_5}{\underline{Y}_4} \right) \underline{Y}_1 + \left(-\underline{V}_A - \underline{V}_A \frac{\underline{Y}_5}{\underline{Y}_4} \right) \underline{Y}_2 - \underline{V}_A \frac{\underline{Y}_5}{\underline{Y}_4} \underline{Y}_3 - \underline{V}_A \frac{\underline{Y}_5}{\underline{Y}_4} \underline{Y}_4, \quad (39)$$

rearranging yields

$$0 = \underline{V}_E \underline{Y}_1 - \underline{V}_A \underline{Y}_2 - \underline{V}_A \frac{\underline{Y}_5}{\underline{Y}_4} (\underline{Y}_1 + \underline{Y}_2 + \underline{Y}_3 + \underline{Y}_4). \quad (40)$$

Finally, this will be rearranged to get the transfer function

$$\underline{H}_{BP}(s) = \frac{\underline{V}_A}{\underline{V}_E} = \frac{\underline{Y}_1 \underline{Y}_4}{\underline{Y}_5 (\underline{Y}_1 + \underline{Y}_2 + \underline{Y}_3 + \underline{Y}_4) + \underline{Y}_2 \underline{Y}_4} \quad (41)$$

Using equation (31), (32) and (33), the transfer function is

$$\underline{H}_{BP}(s) = \frac{7.162 \cdot 10^{-7} \text{sec } s}{1 + 7.162 \cdot 10^{-7} \text{sec } s + 2.533 \cdot 10^{-12} \text{sec}^2 s^2} \quad (42)$$

2.2 Selecting the Components

The transfer function (42) has to be realized by using the transferfunction of the circuit depicted in equation (41). The parts $\underline{Y}_1 \dots \underline{Y}_5$ are selected to:

One s is needed in the numerator, so $\underline{Y}_1 = \frac{1}{R_1}$ and $\underline{Y}_4 = sC_4$ are the first components to be placed. The transferfunction is

$$\underline{H}_{BP}(s) = \frac{\frac{1}{R_1} s C_4}{\underline{Y}_5 \left(\frac{1}{R_1} + \underline{Y}_2 + \underline{Y}_3 + s C_4 \right) + \underline{Y}_2 s C_4}, \quad (43)$$



s^2 is needed in the denominator, so $\underline{Y}_2 = sC_2$ is placed in the transfer function

$$\underline{H}_{BP}(s) = \frac{\frac{1}{R_1} s C_4}{\underline{Y}_5 \left(\frac{1}{R_1} + s C_2 + \underline{Y}_3 + s C_4 \right) + s^2 C_2 C_4}, \quad (44)$$

finally $\underline{Y}_5 = \frac{1}{R_5}$ is selected for a linear term in the denominator. $\underline{Y}_3 = \frac{1}{R_3}$ may be freely selected to be a resistor, too. The transfer function becomes

$$\underline{H}_{BP}(s) = \frac{\frac{1}{R_1} s C_4}{\frac{1}{R_5} \left(\frac{1}{R_1} + s C_2 + \frac{1}{R_3} + s C_4 \right) + s^2 C_2 C_4}. \quad (45)$$

This can be rearranged to

$$\underline{H}_{BP}(s) = \frac{\frac{C_4 R_3 R_5}{R_1 + R_3} s}{1 + \frac{R_1 R_3}{R_1 + R_3} [C_2 + C_4] s + \frac{C_2 C_4 R_1 R_3 R_5}{R_1 + R_3} s^2}. \quad (46)$$

Comparing the coefficients leads to three equations

$$a_1 = 7.162 \cdot 10^{-7} \text{ sec} = \frac{C_4 R_3 R_5}{R_1 + R_3} \quad (47)$$

$$b_1 = 7.162 \cdot 10^{-7} \text{ sec} = \frac{R_1 R_3}{R_1 + R_3} [C_2 + C_4] \quad (48)$$

$$b_2 = 2.533 \cdot 10^{-12} \text{ sec}^2 = \frac{C_2 C_4 R_1 R_3 R_5}{R_1 + R_3} \quad (49)$$

The equation (48) allows to use a R_{eq} which is

$$R_{eq} = \frac{b_1}{C_2 + C_4} = \frac{R_1 R_3}{R_1 + R_3}, \quad (50)$$

which is used to rewrite (49)

$$b_2 = C_2 C_4 R_5 R_{eq}. \quad (51)$$

Now, R_5 can be calculated by

$$R_5 = \frac{b_2}{C_2 C_4 R_{eq}} = \frac{b_2 (C_2 + C_4)}{b_1 C_2 C_4} \quad (52)$$

where $C_2 = C_4 = 1 \text{ nF}$ are set, thus R_5 is

$$R_5 = 7.073 \text{ k}\Omega. \quad (53)$$

From equation (47) follows

$$C_4 R_3 R_5 = a_1 R_1 + a_1 R_3. \quad (54)$$

This is rewritten to

$$R_1 = R_3 \frac{C_4 R_5 - a_1}{a_1}, \quad (55)$$



and inserted in (48)

$$\frac{b_1}{C_2 + C_4} = \frac{R_3^2 \frac{C_4 R_5 - a_1}{a_1}}{R_3 \frac{C_4 R_5 - a_1}{a_1} + R_3} = R_3 \frac{C_4 R_5 - a_1}{C_4 a_1} \quad (56)$$

Finally, R_3 is calculated to

$$R_3 = \frac{C_4 R_5 b_1}{(C_4 R_5 - a_1) \cdot (C_2 + C_4)} = 398.44 \, \Omega, \quad (57)$$

and R_1 by using (55)

$$R_1 = 3.536 \, \text{k}\Omega \quad (58)$$